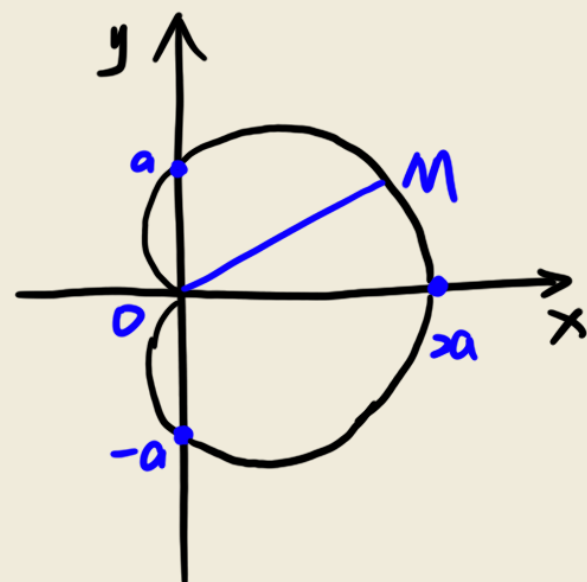


求心脏线 $\rho = a(1 + \cos\theta)$ 围成图形的面积 A

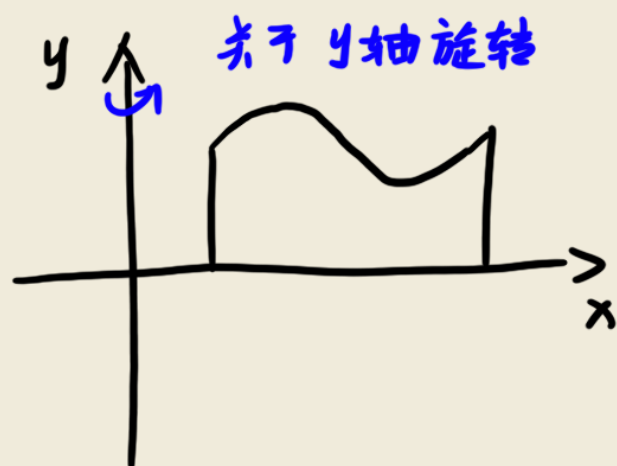


极坐标
 $\Delta A \approx dA = \frac{1}{2} \rho^2(\theta) d\theta$

$$A = \frac{1}{2} \int_{\alpha}^{\beta} \rho^2(\theta) d\theta$$

此处 $A = 2 \cdot \frac{1}{2} \int_0^{\pi} a^2(1 + \cos\theta)^2 d\theta = a^2 \int_0^{\pi} (1 + 2\cos\theta + \frac{1}{2}(1 + \cos 2\theta)) d\theta$
 $= a^2 (\frac{3}{2}\theta + 2\sin\theta + \frac{1}{4}\sin 2\theta) \Big|_0^{\pi} = \frac{3}{2}\pi a^2$

观察出 $\sin\theta$



$$dv = (\pi(x + \Delta x)^2 - \pi x^2) \cdot f(x) = \pi(2x\Delta x + (\Delta x)^2) f(x)$$

$$dv = 2\pi x f(x) dx$$

$$V = \int_a^b 2\pi x f(x) dx$$

$$V_y = 2\pi \int_a^b x f(x) dx$$

例1. 椭圆 $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ ($a, b > 0$) 绕 x 轴旋转和绕 y 轴旋转所得的 V_x, V_y

$$V_x = \pi \int_{-a}^a y^2 dx = 2\pi \int_0^a \frac{b^2}{a^2}(a^2 - x^2) dx = 2\pi b^2(a - \frac{1}{a^2} \cdot \frac{1}{3}a^3) = \frac{4\pi}{3} ab^2$$

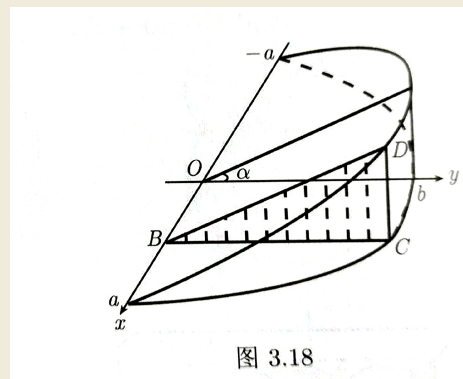
$$V_y = \pi \int_{-b}^b x^2 dy = 2\pi \int_0^b \frac{a^2}{b^2}(b^2 - y^2) dy = 2\pi a^2(b - \frac{1}{b^2} \cdot \frac{1}{3}b^3) = \frac{4\pi}{3} a^2 b$$

例2 $y = \sin x$ ($0 \leq x \leq \pi$) 与 x 轴所围曲边梯形分别绕 x 轴和 y 轴旋转得 V_x, V_y

$$V_x = \pi \int_0^{\pi} \sin^2 x dx = \frac{\pi}{2} \int_0^{\pi} (1 - \cos 2x) dx = \frac{\pi^2}{2}$$

$$V_y = 2\pi \int_0^{\pi} x \sin x dx = -2\pi \int_0^{\pi} x d\cos x = -2\pi x \cos x \Big|_0^{\pi} + 2\pi \int_0^{\pi} \cos x dx = 2\pi^2$$

例3. 求图中楔形的体积



$$y = \frac{b}{a} \sqrt{a^2 - x^2}$$

按 BCD 三角形截面取 $|BC| = \frac{b}{a} \sqrt{a^2 - x^2}$, $|CD| = \frac{b}{a} \sqrt{a^2 - x^2} \tan \alpha$

$$A(x) = \frac{1}{2} \frac{b^2 \tan \alpha}{a^2} (a^2 - x^2)$$

$$V = \int_{-a}^a \frac{b^2 \tan \alpha}{2a^2} (a^2 - x^2) dx = \frac{2}{3} ab^2 \tan \alpha$$

弧长 $M_{i-1} = (x_{i-1}, f(x_{i-1}))$ $M_i = (x_i, f(x_i))$

$$\begin{aligned} |M_{i-1} - M_i| &= \sqrt{(x_i - x_{i-1})^2 + (f(x_i) - f(x_{i-1}))^2} \\ &= \sqrt{1 + \left(\frac{f(x_i) - f(x_{i-1}))}{\Delta x} \right)^2} \Delta x \\ &= \sqrt{1 + f'(x_i)^2} \Delta x \end{aligned}$$

$$S = \int_a^b \sqrt{1 + f'(x)^2} dx$$

参数方程下 $S = \int_a^b \sqrt{1 + \left(\frac{\psi'(t)}{\varphi'(t)} \right)^2} |\varphi'(t)| dt = \int_a^b \sqrt{\varphi'^2(t) + (\psi'(t))^2} dt$

极坐标方程 $\rho = \rho(\theta) \quad \alpha \leq \theta \leq \beta$

$$s = \int_{\alpha}^{\beta} \sqrt{\rho^2(\theta) + (\rho'(\theta))^2} d\theta$$

求心脏线 $\rho = a(1 + \cos\theta)$ 的全长 s

$$ds = \sqrt{\rho^2(\theta) + (\rho'(\theta))^2} d\theta = \sqrt{(a(1 + \cos\theta))^2 + (-a \sin\theta)^2} d\theta = 2a |\cos \frac{\theta}{2}| d\theta$$

$$s = 2 \int_0^{\pi} ds$$

求椭圆 $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \quad (a > b > 0)$ 的周长 s

$$x = a \cos\theta, y = b \sin\theta$$

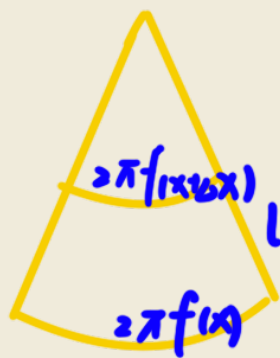
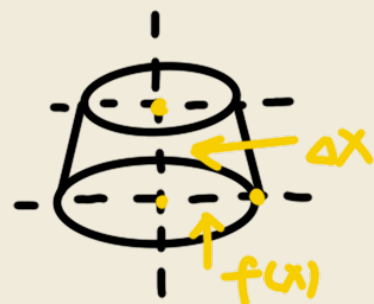
$$ds = \sqrt{(x')^2 + (y')^2} d\theta = \sqrt{a^2 \sin^2\theta + b^2 \cos^2\theta} d\theta$$

$$s = 4 \int_0^{\frac{\pi}{2}} \sqrt{a^2 \sin^2\theta + b^2 \cos^2\theta} d\theta$$

$$\text{令 } e = \frac{\sqrt{a^2 - b^2}}{a} \quad \text{则 } s = 4a \int_0^{\frac{\pi}{2}} \sqrt{1 - e^2 \cos^2\theta} d\theta$$

旋转曲面面积

近似于圆台



$$\sqrt{1+f'(x)^2} \Delta x$$

$$S \text{ 例} \approx 2\pi \int f(x) \sqrt{1+(f'(x))^2} dx$$

$$\Rightarrow A_x = 2\pi \int_a^b f(x) \sqrt{1+(f'(x))^2} dx$$

$$A_y = 2\pi \int_a^b x \sqrt{1+(f'(x))^2} dx$$

$$A_x = 2\pi \int_\alpha^\beta \varphi(t) \sqrt{(\varphi'(t))^2 + (\psi'(t))^2} dt$$

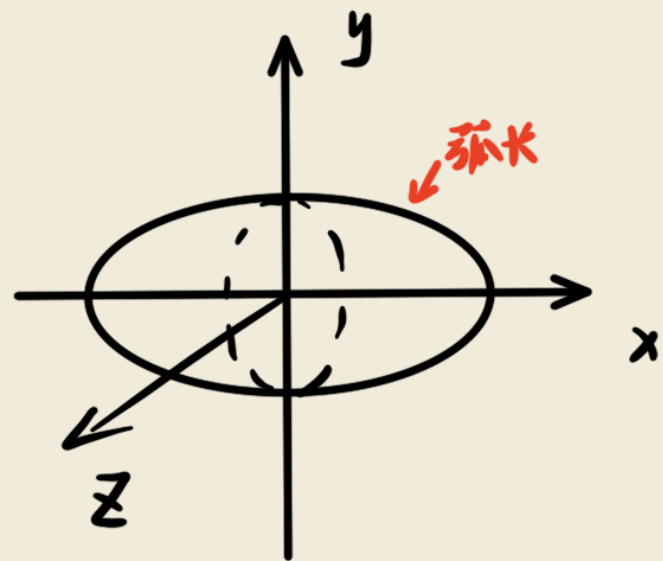
$$A_x = 2\pi \int_\alpha^\beta \rho(\theta) \sin \theta \sqrt{\rho'(\theta)^2 + \rho(\theta)^2} d\theta$$

例. $y = \sqrt{x}$ ($0 \leq x \leq 1$) 绕 x 轴旋转生成的曲面面积

$$dA = 2\pi \sqrt{x} \cdot \sqrt{1 + (\frac{1}{2}\sqrt{x})^2} dx$$

$$A = \int_0^1 2\pi \sqrt{x + \frac{1}{4}} dx = 2\pi \frac{2}{3} (x + \frac{1}{4})^{\frac{3}{2}} \Big|_0^1$$

$$= \frac{4\pi}{3} (\frac{\sqrt{5}}{2} - \frac{1}{2}) = \frac{2\pi(\sqrt{5}-1)}{3}$$



$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{b^2} = 1$$

$$A_x = 2 \int_0^{\frac{\pi}{2}} 2\pi b \sin \theta \sqrt{(a \sin \theta)^2 + (b \cos \theta)^2} d\theta$$

$$= 4\pi b \int_0^{\frac{\pi}{2}} \sqrt{a^2 - (a^2 - b^2) \cos^2 \theta} d(\cos \theta)$$

$$t = \cos \theta \quad 4\pi ab \int_0^1 \sqrt{1 - \frac{a^2 - b^2}{a^2} t^2} dt \quad \frac{\sqrt{a^2 - b^2}}{a^2} \cdot t = \sin \alpha \quad \int_0^{\arcsin \frac{\sqrt{a^2 - b^2}}{a}} \cos^2 \alpha \cdot \frac{a}{\sqrt{a^2 - b^2}} d\alpha$$

$$= \frac{4\pi a^2 b}{\sqrt{a^2 - b^2}} \int_0^{\arcsin \frac{\sqrt{a^2 - b^2}}{a}} \frac{1 + \cos 2\alpha}{2} d\alpha = \frac{4\pi a^2 b}{\sqrt{a^2 - b^2}} \left(\frac{1}{2} \arcsin \frac{\sqrt{a^2 - b^2}}{a} + \frac{b\sqrt{a^2 - b^2}}{2a^2} \right)$$

$$= 2\pi b^2 + \frac{2\pi a^2 b}{\sqrt{a^2 - b^2}} \arcsin \frac{\sqrt{a^2 - b^2}}{a}$$

曲率公式

$$K = \lim_{\Delta x \rightarrow 0} \frac{|\Delta \alpha|}{|\Delta s|} = \lim_{\Delta x \rightarrow 0} \frac{\left| \frac{\Delta \alpha}{\Delta x} \right|}{\left| \frac{\Delta s}{\Delta x} \right|} = \frac{|f''(x_0)|}{(1 + (f'(x_0))^2)^{\frac{3}{2}}}$$

例: 半径为 r 的圆

$$\Delta s = R \cdot \Delta \alpha \Rightarrow K = \frac{1}{R}$$

例: $y = ax^2$ ($a > 0$) 的曲率.

$$K = \frac{2a}{(1 + 4a^2x^2)^{\frac{3}{2}}} \quad x=0 \text{ 时取 } K_{\max} = 2a$$

物理学应用

第一象限内以原点为圆心, 以 R 为半径的均质圆弧质心坐标,

$$\begin{cases} x = R \cos \theta \\ y = R \sin \theta \end{cases} \quad \text{不妨设 } \mu = 1 \text{ 则 } M = \frac{1}{2} \pi R$$

$$\bar{x} = \frac{2}{\pi R} \int_0^{\frac{\pi}{2}} R^2 \cos \theta d\theta = \frac{2R}{\pi}, \quad \bar{y} = \frac{2}{\pi R} \int_0^{\frac{\pi}{2}} R^2 \sin \theta d\theta = \frac{2R}{\pi}$$

求 半径为 R 的半圆的形心 G

$$A = \frac{1}{2} \pi R^2$$

$$\bar{x} = \bar{y} = \frac{1}{A} \int_0^R (R^2 - x^2) dx = \frac{4R}{3\pi}$$

$$G(0, \frac{4R}{3\pi})$$

